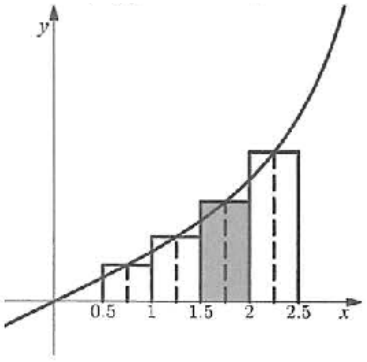
Numerical Methods: Mid-ordinate Rule

Questions 1–19

Question 1. Part of the function $f(x) = \tan 0.4x$ is shown in the diagram below. Four rectangles of equal height have been sketched, each with height determined by $f(x)$ at their midpoints.



Find the area of the shaded rectangle, giving your answer to three significant figures.

Question 2. Using the mid-ordinate rule, calculate an approximation to the integral

$$\int_0^2 e^{-x^2}\,\mathrm{d}x$$

when the width of the strip is given by each of the following values of h, giving your answer to six decimal places:

a) $h = 2$
b) $h = 1$
c) $h = 0.5$

Question 3. Using the mid-ordinate rule, calculate an approximation to the integral

$$\int_1^2 \frac{1}{1-\ln x}\,\mathrm{d}x$$

for each of the following, giving your answer to six decimal places:

a) One strip.
b) Two strips.
c) Four strips.

Question 4.

a) Use the mid-ordinate rule with four strips of equal width to find an estimate for

$$\int_0^2 \frac{\sin x}{x}\,\mathrm{d}x$$

giving your answer to three significant figures.

b) Explain how you could find an improved estimate.

Question 5. The velocity of a car v $\mathrm{m\,s^{-1}}$ at time t seconds is modelled by

$$v = 10 + 3t - 0.2t^2 \quad \text{for } 0 \leq t \leq 10.$$

Use the mid-ordinate rule with five strips to estimate the distance travelled in the first 10 seconds.

Question 6. Prerequisite knowledge for Question 6: Integration of $\ln x$.

a) Use the mid-ordinate rule with three rectangles to find an estimate for

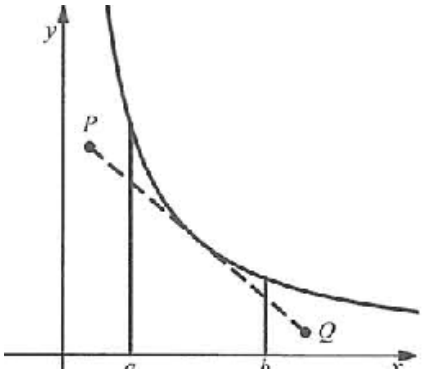
$$I = \int_1^4 \ln x \, \mathrm{d}x$$

giving your answer to four decimal places.

b) Find the exact value of I.

c) Find the relative error of your approximation from part (a).

Question 7. The diagram shows part of the graph of $y = f(x)$. The line PQ lies on the tangent to the graph of $y = f(x)$ at the midpoint between a and b.



a) Use the mid-ordinate rule with one strip to find an expression to estimate the integral $\int_a^b f(x)\,\mathrm{d}x$ in terms of a and b.

b) State, giving reasons, whether you expect your estimate to be an under-estimate or an over-estimate.

Question 8. The graph of $y = f(x)$ is concave (concave downwards) for the interval $[a, b]$. By considering a trapezium touching the curve at the midpoint of a and b, explain why the mid-ordinate rule with one strip overestimates $\int_a^b f(x)\,\mathrm{d}x$.

Question 9.

a) Use the mid-ordinate rule with two strips of equal width to find an estimate for

$$\int_0^1 3\cos x \, \mathrm{d}x$$

giving your answer to two decimal places.

b) Using a sketch of the graph of $y = 3\cos x$ for $0 \leq x \leq 1$, explain clearly whether your answer to part (a) is an overestimate or an underestimate.

Question 10. Prerequisite knowledge for Question 10: Hyperbolic Functions.

a) Use the mid-ordinate rule with four strips of equal width to find an estimate for

$$\int_1^2 \sinh x \, \mathrm{d}x$$

giving your answer to three significant figures.

b) Using a sketch of the graph of $y = \sinh x$ for $1 \leq x \leq 2$, explain clearly whether your answer to part (a) is an overestimate or an underestimate.

Question 11. Prerequisite knowledge for Questions 11–13: Trapezium Rule and the Chain Rule. Let

$$I = \int_1^4 \sqrt{x^2 - 1}\,\mathrm{d}x.$$

a) Find an estimate for I using the trapezium rule with three strips, giving your answer to four decimal places.
b) Find an estimate for I using the mid-ordinate rule with three strips, giving your answer to four decimal places.
Given $f(x) = \sqrt{x^2 - 1}$ and additionally that $f'(x)$ is always decreasing for $1 \leq x \leq 4$,
c) State, giving a reason, whether the trapezium rule gives an over- or under-estimate.
d) State, giving a reason, whether the mid-ordinate rule gives an over- or under-estimate.

Question 12. Let

$$I = \int_2^{10} \frac{1}{\ln x}\,\mathrm{d}x.$$

a) Find an estimate for I using the trapezium rule with four strips, giving your answer to four decimal places.
b) Find an estimate for I using the mid-ordinate rule with four strips, giving your answer to four decimal places.
It is given that $f(x) = \dfrac{1}{\ln x}$.
c) Show that $f'(x)$ is always increasing for $2 \leq x \leq 10$.
d) State, giving a reason, whether the trapezium rule gives an over- or under-estimate.
e) State, giving a reason, whether the mid-ordinate rule gives an over- or under-estimate.

Question 13. Let

$$I = \int_1^3 \frac{e^x}{x}\,\mathrm{d}x.$$

a) Find an estimate for I using the trapezium rule with two strips, giving your answer to five decimal places.

b) Find an estimate for I using the mid-ordinate rule with two strips, giving your answer to five decimal places.

It is known that a better approximation for I is 8.03871 to five decimal places.

c) What does this tell you about the concavity of the curve for $1 \leq x \leq 3$?

Question 14. Prerequisite knowledge for Question 14: Volume of Revolution. A curve has equation $y = \ln(x + 1)$. The region bounded by the curve, the lines $x = 0$ and $x = 2$, and the x-axis is rotated 360° about the x-axis.

a) Write down an expression for the exact volume of revolution.

b) Use the mid-ordinate rule with four strips to estimate this volume, giving your answer to three significant figures.

Question 15. A curve has equation $y = \sin(x^2)$.

a) Use the mid-ordinate rule with two strips to find an estimate for

$$\int_0^1 \sin(x^2)\,\mathrm{d}x$$

giving your answer to four decimal places.

The graph with equation $y = \sin(x^2)$ is stretched with scale factor 3 parallel to the y-axis and then translated by $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$ to give the graph with equation $y = f(x)$.

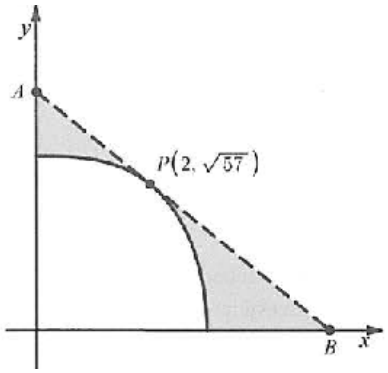
b) Write down an expression for $f(x)$.

Giving your answer to three decimal places:

c) Using your answer to part (a), state an estimate to $\int_2^3 f(x)\,\mathrm{d}x$.

d) Use the mid-ordinate rule with two strips to find an estimate for $\int_2^3 f(x)\,\mathrm{d}x$.

Question 16. The shaded regions on the diagram below are bounded by the curve with equation $y = \sqrt{81 - 3x^3}$, $x \geq 0$, the tangent at P and the coordinate axes. The point P on the curve has coordinates $(2, \sqrt{57})$.



a) Use the mid-ordinate rule with three rectangles to find an estimate for

$$\int_0^3 \sqrt{81 - 3x^3}\,\mathrm{d}x$$

giving your answer to two decimal places.

b) Find the exact form of the equation of the tangent line, giving your answer in the form $ay + b\sqrt{57}x = c\sqrt{57}$, where a, b and c are integers.

c) Hence find an approximate value for the total area of the shaded regions, giving your answer to three significant figures.

Question 17. Prerequisite knowledge for Question 17: Mean Value.

a) Sketch the curve C with equation $y = e^{x/2}$ for $0 \leq x \leq 3$.

The region bounded by the curve C, the coordinate axes and the vertical straight line with equation $x = 3$ is to be estimated by the mid-ordinate rule using three equally spaced strips.

b) Find, correct to six decimal places, the area of this region.

c) Hence estimate the mean value of $y = e^{x/2}$ in the interval $0 \leq x \leq 3$, giving your answer to five decimal places.

Question 18. Prerequisite knowledge for Questions 18–19: Maclaurin Series.

a) State the first three non-zero terms in the Maclaurin expansion of $\ln(1 + x)$.

b) Use the mid-ordinate rule with four strips and this expansion to estimate

$$\int_{0.1}^{0.5} \ln(1 + x)\,\mathrm{d}x$$

giving your answer as a simplified fraction.

Question 19.

a) Find the first three non-zero terms in the Maclaurin expansion of xe^x.

b) Use the mid-ordinate rule with six strips and this expansion to estimate

$$\int_0^{0.3} xe^x \, \mathrm{d}x$$

giving your answer to four decimal places.